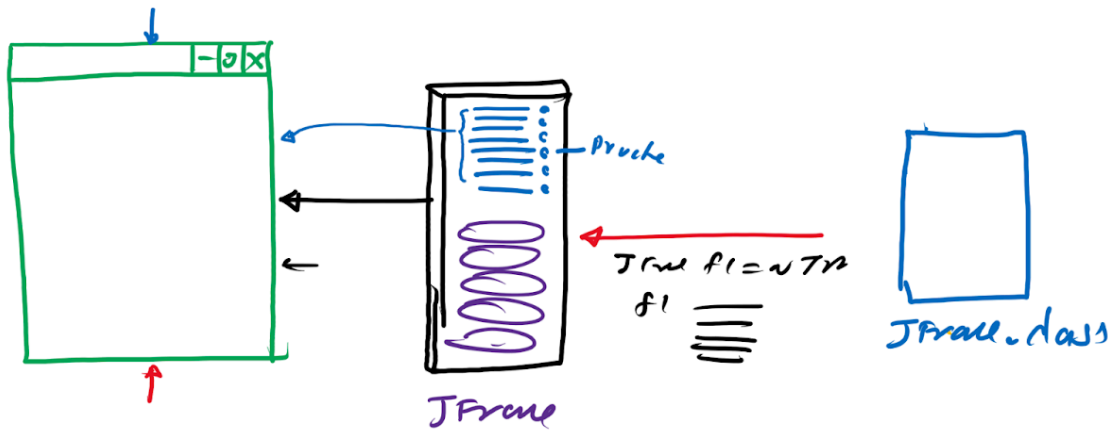


Java Swing Applications (GUI)



Java Swing App.

1. Container (JFrame, JWindow, JPanel, JDialog, -----)
2. Controller (JButton, JTextLabel, JTextArea, JList -----)
3. Layout Manager (BorderLayout, FlowLayout, GridLayout, ---)
4. Event Handling

Q132

```
import javax.swing.*;

class Demo{
    public static void main(String args[]){
        JFrame f1=new JFrame();
        f1.setSize(300,300);
        f1.setTitle("Calculator");
        f1.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        f1.setLocationRelativeTo(null);
        f1.setVisible(true);
    }
}
```

```
import javax.swing.*;

class Demo{

    public static void main(String args[]){

        JFrame f1=new JFrame();

        f1.setSize(300,300);

        f1.setTitle("Calculator");

        f1.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        f1.setLocationRelativeTo(null);


        JButton b1=new JButton();

        b1.setText("Exit");

        f1.add(b1);


        JTextField t1=new JTextField(5);

        f1.add(t1);


        f1.setVisible(true);

    }

}
```

Layout Manager (BorderLayout)

Q135

```
import javax.swing.*;
```

```
class Demo{
```

```
public static void main(String args[]){
```

```
JFrame f1=new JFrame();
```

```
f1.setSize(300,300);
```

```
f1.setTitle("Calculator");
```

```
f1.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
```

```
f1.setLocationRelativeTo(null);
```

```
//Default Layout "BorderLayout"
```

```
JButton b1=new JButton();
```

```
b1.setText("Exit");
```

```
f1.add("South",b1);
```

```

JTextField t1=new JTextField(5);

```

```
f1.add("North",t1);
```

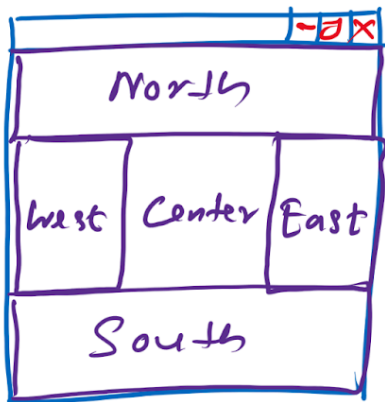
```
f1.setVisible(true);
```

}

}

////////////////////////////////////

=====



South
North
West
East
Center.

} J Buttons

```
class Demo{
    public static void main(String args[]){
        JFrame f1=new JFrame();
        f1.setSize(300,300);
        f1.setTitle("Calculator");
        f1.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        f1.setLocationRelativeTo(null);

        //Default Layout "BorderLayout"
        JButton btnCenter=new JButton("Center");
        JButton btnSouth=new JButton("South");
        JButton btnNorth=new JButton("North");
        JButton btnWest=new JButton("West");
        JButton btnEast=new JButton("East");

        btnCenter.setFont(new Font("",Font.BOLD,25));
        btnSouth.setFont(new Font("",Font.BOLD,25));
        btnNorth.setFont(new Font("",Font.BOLD,25));
        btnWest.setFont(new Font("",Font.BOLD,25));
        btnEast.setFont(new Font("",Font.BOLD,25));

        f1.add("Center",btnCenter);
        f1.add("South",btnSouth);
        f1.add("North",btnNorth);
        f1.add("West",btnWest);
        f1.add("East",btnEast);

        f1.pack();

        f1.setVisible(true);
    }
}
```

=====

```
class Demo{
    public static void main(String args[]){
        MyFrame f1=new MyFrame();
        f1.setSize(300,300);
        f1.setTitle("My Demo Frame");
        f1.setVisible(true);
    }
}
```



```
import javax.swing.*;
import java.awt.*;

class Demo{
    public static void main(String args[]){
        MyFrame f1=new MyFrame();
        f1.setSize(300,300);
        f1.setTitle("My Demo Frame");
        f1.setVisible(true);
    }
}
```

Q139 Step I

=====

```
class MyFrame{
    //
    //
}
```

```
class Demo{  
    public static void main(String args[]){  
        MyFrame f1=new MyFrame();  
        f1.setSize(300,300);  
        f1.setTitle("My Demo Frame");  
        f1.setDefaultCloseOperation(MyFrame.EXIT_ON_CLOSE);  
        f1.setVisible(true);  
    }  
}
```

////////////////////////////////////

Q140 Step II

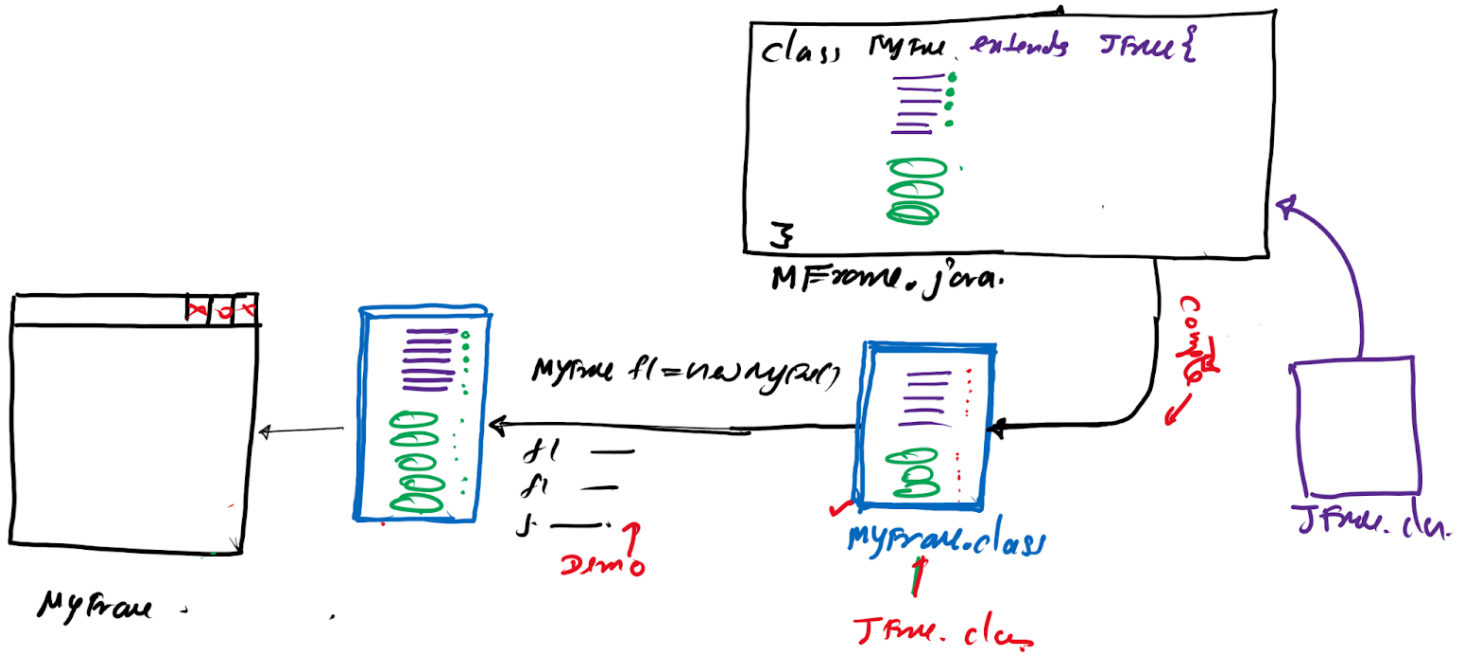
=====

```
import javax.swing.*;  
import java.awt.*;
```

```
class MyFrame extends JFrame{  
    //  
    //  
}
```

```
class Demo{  
    public static void main(String args[]){  
        MyFrame f1=new MyFrame();  
        f1.setSize(300,300);  
        f1.setTitle("My Demo Frame");  
        f1.setDefaultCloseOperation(MyFrame.EXIT_ON_CLOSE);  
        f1.setVisible(true);  
    }  
}
```

```
//MyFrame is a JFrame  
//JFrame --> Super class  
//MyFrame -> Subclass
```



```
class Demo{
    public static void main(String args[]){
        new MyFrame().setVisible(true);
    }
}
```

```
class MyFrame extends JFrame{
    private JButton btnCenter;
    private JButton btnSouth;
    private JButton btnNorth;
    private JButton btnWest;
    private JButton btnEast;

    MyFrame(){
        setSize(300,300);
        setTitle("My Demo Frame");
        setDefaultCloseOperation(EXIT_ON_CLOSE);
        setLocationRelativeTo(null);

        btnCenter=new JButton("Center");
        btnCenter.setFont(new Font("",1,25));
        add("Center",btnCenter);

        btnSouth=new JButton("South");
        btnSouth.setFont(new Font("",1,25));
        add("South",btnSouth);

        btnNorth=new JButton("North");
        btnNorth.setFont(new Font("",1,25));
        add("North",btnNorth);

        btnWest=new JButton("West");
        btnWest.setFont(new Font("",1,25));
        add("West",btnWest);

        btnEast=new JButton("East");
        btnEast.setFont(new Font("",1,25));
        add("East",btnEast);

    }
}

class Demo{
    public static void main(String args[]){
        new MyFrame().setVisible(true);
    }
}
```

Layout manager - "FlowLayout"

Q144

```
import javax.swing.*;
import java.awt.*;

class MyFrame extends JFrame{
    private JButton btnCenter;
    private JButton btnSouth;
    private JButton btnNorth;
    private JButton btnWest;
    private JButton btnEast;

    MyFrame(){
        setSize(300,300);
        setTitle("My Demo Frame");
        setDefaultCloseOperation(EXIT_ON_CLOSE);
        setLocationRelativeTo(null);

        setLayout(new FlowLayout());

        btnCenter=new JButton("Center");
        btnCenter.setFont(new Font("",1,25));
        add(btnCenter);

        btnSouth=new JButton("South");
        btnSouth.setFont(new Font("",1,25));
        add(btnSouth);

        btnNorth=new JButton("North");
        btnNorth.setFont(new Font("",1,25));
        add(btnNorth);

        btnWest=new JButton("West");
        btnWest.setFont(new Font("",1,25));
        add(btnWest);

        btnEast=new JButton("East");
        btnEast.setFont(new Font("",1,25));
        add(btnEast);

    }
}
```

////////////////////////////////////

```
import javax.swing.*.*;
import java.awt.*.*;
```

```
class MyFrame extends JFrame{
    private JButton btnCenter;
    private JButton btnSouth;
    private JButton btnNorth;
    private JButton btnWest;
    private JButton btnEast;

    MyFrame(){
        setSize(300,300);
        setTitle("My Demo Frame");
        setDefaultCloseOperation(EXIT_ON_CLOSE);
        setLocationRelativeTo(null);

        setLayout(new FlowLayout(FlowLayout.RIGHT)); //FlowLayout.LEFT
        //FlowLayout.CENTER
        btnCenter=new JButton("Center");
        btnCenter.setFont(new Font("",1,25));
        add(btnCenter);

        btnSouth=new JButton("South");
        btnSouth.setFont(new Font("",1,25));
        add(btnSouth);

        btnNorth=new JButton("North");
        btnNorth.setFont(new Font("",1,25));
        add(btnNorth);

        btnWest=new JButton("West");
        btnWest.setFont(new Font("",1,25));
        add(btnWest);

        btnEast=new JButton("East");
        btnEast.setFont(new Font("",1,25));
        add(btnEast);
    }
}
```

////////////////////////////////////

new JTextField(s)

BorderLayout

FlowLayout

////////////////////////////////////

```
class Demo{
    public static void main(String args[]){
        new MyFrame().setVisible(true);
    }
}
```

```
f1.setLayout(new GridLayout(3, 4))
```

Q148

```
class MyFrame extends JFrame{
    private JButton btnCenter;
    private JButton btnSouth;
    private JButton btnNorth;
    private JButton btnWest;

    MyFrame(){
        setSize(300,300);
        setTitle("My Demo Frame");
        setDefaultCloseOperation(EXIT_ON_CLOSE);
        setLocationRelativeTo(null);

        setLayout(new GridLayout(2,2));

        btnCenter=new JButton("Center");
        btnCenter.setFont(new Font("",1,25));
        add(btnCenter);

        btnSouth=new JButton("South");
        btnSouth.setFont(new Font("",1,25));
        add(btnSouth);

        btnNorth=new JButton("North");
        btnNorth.setFont(new Font("",1,25));
        add(btnNorth);

        btnWest=new JButton("West");
        btnWest.setFont(new Font("",1,25));
        add(btnWest);

    }
}

class Demo{
    public static void main(String args[]){
        new MyFrame().setVisible(true);
    }
}
```

Q149 Exercise

=====



Q150 From Q149

=====

```
import javax.swing.*;
import java.awt.*;
```

```
class MyFrame extends JFrame{
    private JButton[] btnArray;

    MyFrame(){
        setSize(300,300);
        setTitle("My Demo Frame");
        setDefaultCloseOperation(EXIT_ON_CLOSE);
        setLocationRelativeTo(null);

        setLayout(new GridLayout(4,4,2,2));

        btnArray=new JButton[16];
        String[] btnNames={"7","8","9","*","4","5","6","/","1","2","3","+","0",".","=","-"};
        for (int i = 0; i < 16; i++){
            btnArray[i]=new JButton(btnNames[i]);
            btnArray[i].setFont(new Font("",1,20));
            add(btnArray[i]);
        }
    }
}

class Demo{
    public static void main(String args[]){
        new MyFrame().setVisible(true);
    }
}
```

////////////////////////////////////
Mixed Layout
////////////////////////////////////

Q151

```
import javax.swing.*;
```

```
import java.awt.*;
```

```
class MyFrame extends JFrame{
    private JButton btnCenter;
    private JButton btnSouth;
    private JButton btnNorth;
    private JButton btnWest;

    private JTextField textField;
    MyFrame(){
        setSize(300,300);
        setTitle("My Demo Frame");
        setDefaultCloseOperation(EXIT_ON_CLOSE);
        setLocationRelativeTo(null);

        JPanel centerPanel=new JPanel();
        centerPanel.setLayout(new GridLayout(2,2));

        btnCenter=new JButton("Center");
        btnCenter.setFont(new Font("",1,25));
        centerPanel.add(btnCenter);

        btnSouth=new JButton("South");
        btnSouth.setFont(new Font("",1,25));
        centerPanel.add(btnSouth);

        btnNorth=new JButton("North");
        btnNorth.setFont(new Font("",1,25));
        centerPanel.add(btnNorth);

        btnWest=new JButton("West");
        btnWest.setFont(new Font("",1,25));
        centerPanel.add(btnWest);

        add("Center",centerPanel);

        textField=new JTextField(5);
        textField.setFont(new Font("",1,20));
        add("North",textField);
    }
}
```

////////////////////////////////////

=====



=====

```
class MyFrame extends JFrame{
    private JButton[] btnArray;
    private JTextField textField;
    MyFrame(){
        setSize(300,300);
        setTitle("My Demo Frame");
        setDefaultCloseOperation(EXIT_ON_CLOSE);
        setLocationRelativeTo(null);

        textField=new JTextField(5);
        textField.setFont(new Font("",1,20));
        add("North",textField);

        JPanel buttonPanel=new JPanel();
        buttonPanel.setLayout(new GridLayout(4,4,2,2));

        btnArray=new JButton[16];
        String[] btnNames={"7","8","9","*","4","5","6","/","1","2","3","+","0",".", "=", "-"};
        for (int i = 0; i < 16; i++){
            btnArray[i]=new JButton(btnNames[i]);
            btnArray[i].setFont(new Font("",1,20));
            buttonPanel.add(btnArray[i]);
        }
        add("Center",buttonPanel);
    }
}
```

```

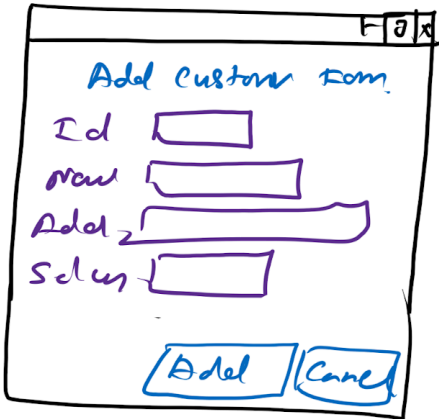
class Demo{
    public static void main(String args[]){
        new MyFrame().setVisible(true);
    }
}

```

```

////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
AddCustomerForm

```



```

////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
Q154 Step I
=====

```

```

import javax.swing.*;
import java.awt.*;

class AddCustomerForm extends JFrame{
    AddCustomerForm(){
        setTitle("Add Customer Form");
        setSize(600,400);
        setDefaultCloseOperation(EXIT_ON_CLOSE);
        setLocationRelativeTo(null);
    }
}

class Demo{
    public static void main(String args[]){
        new AddCustomerForm().setVisible(true);
    }
}

```

```

////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////////
Q155 Step II
=====

```

```

import javax.swing.*;
import java.awt.*;

class AddCustomerForm extends JFrame{
    private JLabel titleLabel;

```



```
JPanel buttonPanel=new JPanel(new FlowLayout(FlowLayout.RIGHT));
btnAdd=new JButton("Add Customer");
btnAdd.setFont(new Font("",1,20));
buttonPanel.add(btnAdd);
```

```
btnCancel=new JButton("Cancel");
btnCancel.setFont(new Font("",1,20));
buttonPanel.add(btnCancel);
```

```
add("South",buttonPanel);
//-----
```

```
JPanel centerPanel=new JPanel(new GridLayout(4,2));
```

```
lblId=new JLabel("Customer Id");
lblId.setFont(new Font("",1,20));
centerPanel.add(lblId);
txtId=new JTextField(5);
txtId.setFont(new Font("",1,20));
centerPanel.add(txtId);
```

```
lblName=new JLabel("Name");
lblName.setFont(new Font("",1,20));
centerPanel.add(lblName);
txtName=new JTextField(10);
txtName.setFont(new Font("",1,20));
centerPanel.add(txtName);
```

```
lblAddress=new JLabel("Address");
lblAddress.setFont(new Font("",1,20));
centerPanel.add(lblAddress);
txtAddress=new JTextField(20);
txtAddress.setFont(new Font("",1,20));
centerPanel.add(txtAddress);
```

```
lblSalary=new JLabel("Salary");
lblSalary.setFont(new Font("",1,20));
centerPanel.add(lblSalary);
txtSalary=new JTextField(6);
txtSalary.setFont(new Font("",1,20));
centerPanel.add(txtSalary);
```

```
add("Center",centerPanel);
pack();
```

```
}
```

```
}
```



```

add("South",buttonPanel);
//-----

JPanel labelPanel=new JPanel(new GridLayout(4,1));
lblId=new JLabel("Customer Id");
lblId.setFont(new Font("",1,20));
labelPanel.add(lblId);
lblName=new JLabel("Name");
lblName.setFont(new Font("",1,20));
labelPanel.add(lblName);
lblAddress=new JLabel("Address");
lblAddress.setFont(new Font("",1,20));
labelPanel.add(lblAddress);
lblSalary=new JLabel("Salary");
lblSalary.setFont(new Font("",1,20));
labelPanel.add(lblSalary);
add("West",labelPanel);

```

```

JPanel textPanel=new JPanel(new GridLayout(4,1));

```

```

txtId=new JTextField(5);
txtId.setFont(new Font("",1,20));
textPanel.add(txtId);

```

```

txtName=new JTextField(10);
txtName.setFont(new Font("",1,20));
textPanel.add(txtName);

```

```

txtAddress=new JTextField(20);
txtAddress.setFont(new Font("",1,20));
textPanel.add(txtAddress);

```

```

txtSalary=new JTextField(6);
txtSalary.setFont(new Font("",1,20));
textPanel.add(txtSalary);

```

```

add("East",textPanel);
pack();

```

```

}
}
class Demo{
    public static void main(String args[]){
        new AddCustomerForm().setVisible(true);
    }
}

```

```
JPanel labelPanel=new JPanel(new GridLayout(4,1));
lblId=new JLabel("Customer Id");
```

```

lblId.setFont(new Font("",1,20));
labelPanel.add(lblId);

lblName=new JLabel("Name");
lblName.setFont(new Font("",1,20));
labelPanel.add(lblName);
lblAddress=new JLabel("Address");
lblAddress.setFont(new Font("",1,20));
labelPanel.add(lblAddress);
lblSalary=new JLabel("Salary");
lblSalary.setFont(new Font("",1,20));
labelPanel.add(lblSalary);
add("West",labelPanel);

JPanel textPanel=new JPanel(new GridLayout(4,1));

txtId=new JTextField(5);
txtId.setFont(new Font("",1,20));
JPanel idTextPanel=new JPanel(new FlowLayout(FlowLayout.LEFT));
idTextPanel.add(txtId);
textPanel.add(idTextPanel);

txtName=new JTextField(10);
txtName.setFont(new Font("",1,20));
textPanel.add(txtName);

txtAddress=new JTextField(20);
txtAddress.setFont(new Font("",1,20));
textPanel.add(txtAddress);

txtSalary=new JTextField(6);
txtSalary.setFont(new Font("",1,20));
textPanel.add(txtSalary);

add("East",textPanel);
pack();
}
}
class Demo{
    public static void main(String args[]){
        new AddCustomerForm().setVisible(true);
    }
}

```

Exercise- Notepad

Q160 Step I

```
import javax.swing.*;
```

```
import java.awt.*;
```

```
class Notepad extends JFrame{
```

```
Notepad(){
    setTitle("Notepad");
    setSize(600,400);
    setDefaultCloseOperation(EXIT_ON_CLOSE);
    setLocationRelativeTo(null);
}
```

}

```
class Demo{
    public static void main(String args[]){
        new Notepad().setVisible(true);
    }
}
```

Q161 Step II

```
import javax.swing.*;
```

```
import java.awt.*;
```

```
class Notepad extends JFrame{
```

```
private JTextArea textArea;
```

```
Notepad(){  
    setTitle("Notepad");  
    setSize(600,400);  
    setDefaultCloseOperation(EXIT_ON_CLOSE);  
    setLocationRelativeTo(null);  
  
    textArea=new JTextArea();  
    textArea.setFont(new Font("",1,20));  
    add("Center",textArea);  
}
```

}

```
class Demo{
    public static void main(String args[]){
        new Notepad().setVisible(true);
    }
}
```

```

////////////////////////////////////////////////////////////////////////////////////////////////
Q162 Step III
=====
import javax.swing.*;
import java.awt.*;

class Notepad extends JFrame{
    private JTextArea textArea;

    Notepad(){
        setTitle("Notepad");
        setSize(600,400);
        setDefaultCloseOperation(EXIT_ON_CLOSE);
        setLocationRelativeTo(null);

        textArea=new JTextArea();
        textArea.setFont(new Font("",1,20));
        JScrollPane textPane=new JScrollPane(textArea);
        add("Center",textPane);
    }
}

class Demo{
    public static void main(String args[]){
        new Notepad().setVisible(true);
    }
}

```



```
class Demo{
    public static void main(String args[]){
        new MyFrame().setVisible(true);
    }
}
```

```
String[] columnName={"Customer ID", "Name", "Address","Salary"};
dtm=new DefaultTableModel(columnName,0);
```

```

tblCustomerDetails=new JTable(dtm);

JScrollPane tablePane=new JScrollPane(tblCustomerDetails);
add("Center",tablePane);

btnAddCustomer=new JButton("Add Customer");
btnAddCustomer.setFont(new Font("",1,20));

btnAddCustomer.addActionListener(new ActionListener(){
    public void actionPerformed(ActionEvent evt){
        Scanner input=new Scanner(System.in);
        System.out.print("Input customer id : ");
        String id=input.nextLine();

        System.out.print("Input customer name : ");
        String name=input.nextLine();

        System.out.print("Input customer address : ");
        String address=input.nextLine();

        System.out.print("Input salary : ");
        double salary=input.nextDouble();

        Object[] rowData={id,name,address,salary};
        dtm.addRow(rowData);
    }
});

JPanel buttonPanel=new JPanel();
buttonPanel.add(btnAddCustomer);
add("South",buttonPanel);
}
}

class Demo{
    public static void main(String args[]){
        new ViewCustomerForm().setVisible(true);
    }
}

```